

Fortifying human milk for preterm babies: what the evidence supports, and what to tell parents

Fortification of expressed breast milk is routine practice on almost every neonatal unit, yet the evidence behind *when*, *how much*, and *which product* is thinner — and in places more contested — than daily practice suggests. This article sets out what nurses and the wider neonatal team can say with confidence, what remains genuinely uncertain, and how to talk to parents about both.

Key points

- Fortification gives a modest, well-replicated boost to short-term growth, but no trial of any fortification variable has yet shown a clear benefit to long-term neurodevelopment.
- The evidence on exactly when to start fortifying does not favour starting earlier; one meta-analysis links early fortification to a longer hospital stay.
- For babies with an abnormal antenatal Doppler scan, and for babies restarting feeds after surgical NEC or SIP, there is no fortification-specific trial evidence at all — current practice in these groups is extrapolated, not evidence-based.
- Whether human-milk-based fortifier reduces NEC or death compared with cows'-milk-based fortifier is unresolved: several meta-analyses suggest benefit, but the largest single trial found none.
- Parents can be told plainly that fortification helps babies grow a little faster in hospital, and that this has not been shown to change how they develop later — a message that is often more reassuring than clinicians expect.

Why this matters

Babies born preterm and fed on mother's own or donor milk accumulate a nutrient shortfall, because mature breast milk was never designed to match the nutrient demands of a baby who should still be *in utero*. Multi-nutrient fortifiers — adding protein, energy, calcium and phosphate to expressed milk — are added on the reasonable premise that closing this gap will help babies grow and, eventually, develop better. That premise turns out to be only partly supported by trial evidence, and the gaps are not evenly spread: they cluster in exactly the babies where families most want a clear answer — those with growth restriction, and those recovering from bowel disease.

This article draws on 50 studies verified against PubMed, spanning eight questions: does fortification work, when should it start, is it safe in high-risk babies, what happens after NEC, which product is best, does it upset babies' tummies, does it help brains, and what happens after discharge. Practice already varies between UK units on several of these questions — not because anyone is doing something unreasonable, but because the trial evidence genuinely does not point clearly in one direction. Naming that variation, rather than assuming a "right answer" exists somewhere, is the more honest starting point for both team discussion and parent conversations.

Does fortification actually help growth?

Yes, modestly and consistently. The current Cochrane review of 18 trials (1456 babies) found low-to-moderate certainty evidence that fortification increases weight gain by around 1.8 g/kg/day, linear growth by about 0.1 cm/week, and head growth by about 0.06 cm/week, with no clear effect on NEC risk in either direction ^[1]. A 2026 update using a newer trial-quality screening tool found larger benefits from 15 trials, but its own authors rated the evidence low certainty because most of the included trials raised trustworthiness concerns ^[2]. A large network analysis of 59 studies found that combining mother's milk, donor milk and a human-milk-based fortifier ranked best for survival and time to full feeds — but concluded there still isn't enough good evidence to recommend one

feeding regimen over another with confidence ^[3].

Practice point: it is fair to tell colleagues, and parents, that fortification reliably helps growth in hospital by a small amount. It is not yet fair to say it improves long-term outcomes (see below). Growth velocity is a *surrogate* measure — it is easy to record and correlates, at a population level, with later development, but a surrogate is not the same as the outcome itself. A treatment can move a surrogate measure in the "right" direction without moving the outcome families actually care about, and that is precisely the pattern seen once fortification trials look at brain development directly rather than at growth charts.

Question	What the evidence shows	Certainty	Ref
Does fortification help growth?	Modest, consistent increase in weight/length/head-growth velocity across large reviews	Low–Moderate	1,2
Does timing (when to start) matter?	No consistent advantage to starting earlier; one meta-analysis links early start to longer hospital stay	Low–Very low	4–6
Is it safe in Doppler-abnormal/growth-restricted babies?	No fortification-specific trial exists; practice is extrapolated from feed-initiation trials	Not evaluable	10–11
When to restart after NEC/SIP?	Some evidence for restarting feeds; no evidence at all for fortifier-specific timing	Very low / absent	15–18
Does fortifier source (cow's milk vs human milk) matter?	Smaller meta-analyses suggest benefit for human-milk-based; largest single trial found no difference	Low–Moderate, inconsistent	19,22
Does it upset babies' tummies?	Stool hardening tracks with fortifier dose regardless of product; early/fast/full-strength introduction raises intolerance risk	Low–Very low	31,32
Does it help the brain long-term?	No trial of any fortification variable has shown a neurodevelopmental benefit	Moderate (for the null)	23,36
Does continuing it at home help?	No growth/IQ benefit shown; may help sustain breastfeeding instead	Low–Moderate	39–41

Table 1. What the evidence shows, by clinical question. Reference numbers match the main text; full citations in the separate References document.

Getting the timing right

Two different questions get asked here, and they have different answers: *how much milk* a baby should be tolerating before fortifier starts, and *how many days old* a baby should be. The Cochrane review of early-versus-late fortification pooled only two trials (237 babies) comparing starting fortifier at 20–40 mL/kg/day feeds against the conventional ~100 mL/kg/day, and found no difference in growth, tolerance or NEC ^[4]. An earlier systematic review reached the same conclusion from the same two trials ^[5]. A separate meta-analysis of three trials (309 babies) found no growth benefit from starting early, but a **significantly longer hospital stay** with early fortification, alongside a non-significant trend toward more feed intolerance and NEC ^[6]. The original 20-versus-100 mL/kg/day trial itself found no difference in days to full feeds ^[7].

Newer trials look instead at postnatal day. In extremely preterm babies (≤ 28 weeks), starting a human-milk-based fortifier from day 2 did not improve the trial's main body-composition outcome, though babies grew slightly longer and had smaller head-growth faltering ^[8]. A second trial comparing day 4–7 against day 10–14 was similarly neutral on its main outcome, with secondary gains in weight and length favouring the earlier start ^[9].

Practice point: there is no trial evidence that starting fortifier earlier than the conventional threshold improves growth or safety, and one study suggests it may lengthen admission. This is a reasonable, evidence-consistent thing to say if a unit is asked why it hasn't adopted very-early fortification.

Babies with growth restriction or an abnormal Doppler scan

This is the largest evidence gap in the whole topic, and it is worth being explicit about it rather than letting practice imply otherwise. No trial has ever tested fortification — timing, dose, or safety — specifically in babies with antenatal absent or reversed end-diastolic flow (AEDF/REDF) on umbilical artery Doppler. What exists instead is a trial about *starting feeds*, not fortifying them: the ADEPT trial (404 babies) found that starting enteral feeds on day 2 rather than day 6 in growth-restricted, Doppler-abnormal babies got them to full feeds faster with no increase in NEC ^[10]. A planned subgroup of the same trial found babies below 29 weeks tolerated the identical feeding plan far worse than those 29 weeks or older — 39% developed NEC compared with 10% — which is itself a warning against assuming one rule fits this whole group, let alone extending it to fortification ^[11].

An Italian cohort of 570 babies confirmed a clear risk gradient by Doppler pattern — the more severe the Doppler abnormality, the higher the rates of feeding intolerance, NEC and SIP — but this study did not look at fortification at all ^[12]. It is also worth separating two groups that get talked about together: babies with an abnormal antenatal Doppler scan (AEDF/REDF) are a smaller, biologically distinct group from the broader category of babies who are simply small for gestational age with a normal Doppler. The evidence gap described here applies most directly to the Doppler-abnormal group; babies who are small but had a normal Doppler are usually managed as an extension of general preterm/very-low-birthweight practice, though a dedicated trial in this group does not exist either. The biological rationale for caution goes back to two small, decades-old UK studies showing markedly higher NEC rates in babies with AREDF regardless of prematurity or growth restriction ^[13,14] — important for understanding *why* clinicians are cautious, but not fortification evidence.

Communicating with parents: families of growth-restricted or Doppler-flagged babies often ask, reasonably, why fortification is being delayed. It is honest and appropriate to say: *"We wait until your baby is tolerating full feeds well because babies like yours can have a more sensitive tummy — we don't yet have research telling us the best time to add extra nutrients in this situation specifically, so we're being cautious based on what we know about feeding generally."* This avoids implying a certainty that does not exist, while still explaining the rationale for caution.

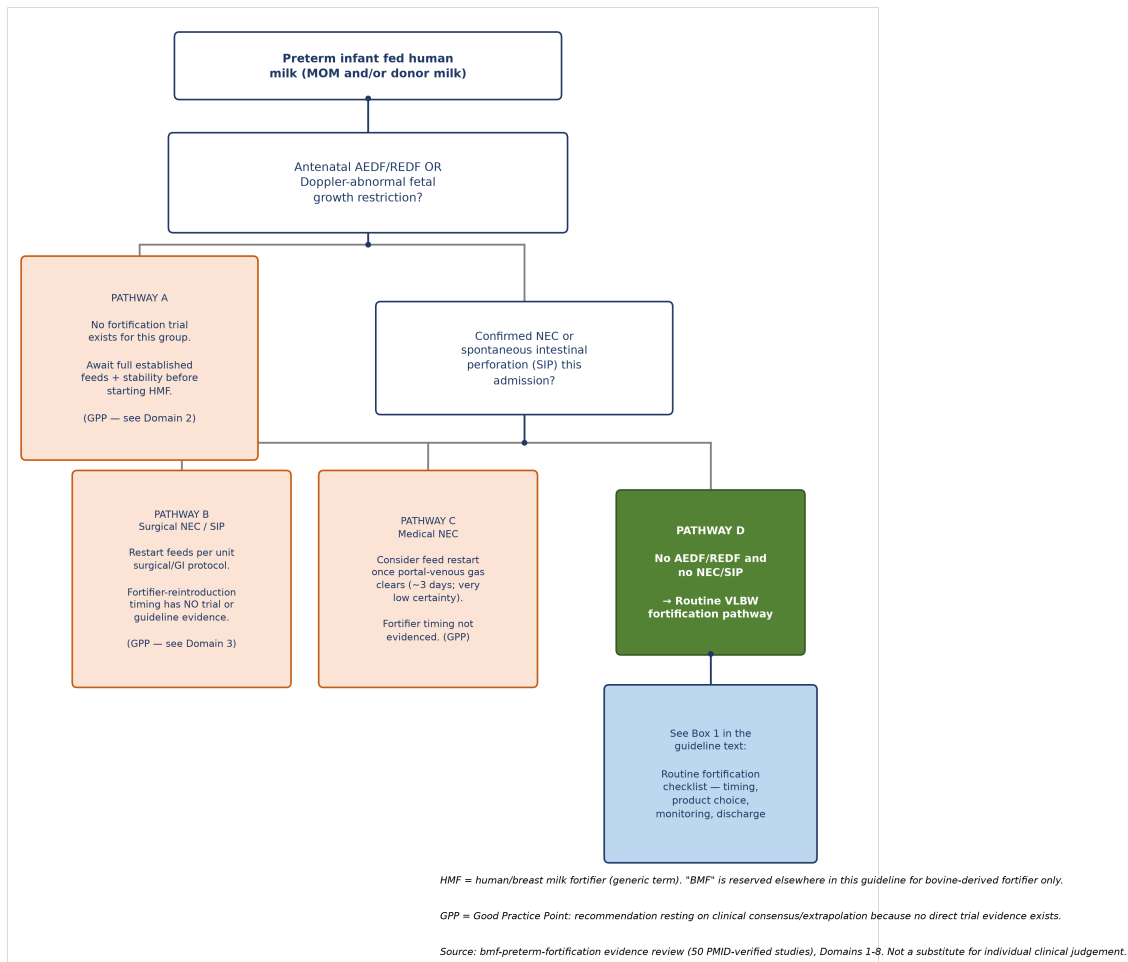


Figure 1. Triage pathway for fortification decisions, from the companion unit guideline built on the same evidence base.

After necrotising enterocolitis

Restarting *any* feed after NEC has some evidence behind it; restarting *fortifier* afterwards does not. A German cohort found that restarting feeds once portal-venous gas had cleared on ultrasound for three days — a median of 4 days after medical NEC — got babies to full feeds faster with shorter line time and less catheter sepsis, without a clear rise in NEC recurrence, though the authors themselves said their study was too small to rule out a higher recurrence risk [15]. A systematic review of feeding after spontaneous intestinal perforation, drawing mostly on older historical-comparison studies, found earlier enteral feeding after SIP was linked to less SIP and, in a couple of studies, less death or disability [16].

Neither the 2024 European surgical NEC guideline nor a 2026 joint Italian neonatal/surgical/nutrition position paper — both checked specifically for this article — contain any recommendation about when to reintroduce fortifier after surgical NEC or SIP [17,18]. This silence, from two current multidisciplinary guideline efforts, is the finding: whatever a unit does here is clinical judgement, not guideline-based practice.

Communicating with parents: after NEC or SIP, parents are often anxious about *any* change to feeding, fortifier included. It helps to separate the two messages clearly: *"We have some evidence about the best way to restart milk feeds after this kind of bowel problem. We don't have specific research on exactly when to add the fortifier back in, so your team will judge that step-by-step based on how your baby's tummy is coping."*

Choosing a fortifier: cows'-milk or human-milk-based?

This is the single most inconsistent question in the evidence, and both sides of the inconsistency deserve airtime. Three meta-analyses point toward a benefit for human-milk-based fortifier: one pooling six trials found lower NEC and surgical NEC with human-milk-based fortifier, but also *lower* weight gain than with cows'-milk-based fortifier — and its NEC benefit lost statistical significance once poorer-quality trials were

excluded ^[19]. A related two-trial analysis found a similar NEC-reduction direction ^[20]. A four-trial analysis restricted to extremely preterm babies found a significant reduction in death with human-milk-based fortifier ^[21].

Set against all three: the Swedish N-forte trial, the largest and most carefully conducted single trial of this exact comparison (228 babies), found **no difference** in the combined outcome of NEC, sepsis or death between human-milk-based and cows'-milk-based fortifier ^[22]. Its companion 18-month follow-up in a separate cohort found no difference in cognitive, language or motor development between fortifier sources either ^[23], and the original trial behind that follow-up found no difference in feeding tolerance ^[24].

Other product questions have clearer single-trial answers. Where a non-acidified liquid fortifier is available, it is probably worth preferring: one adequately sized trial found substantially more metabolic acidosis with an acidified liquid product than a non-acidified one (27% versus 5% of babies), plus more vomiting and gastric residuals ^[25]. Concentrated liquid preterm formula used as a fortifier has shown good early results as a powder alternative ^[26,27], and cows'-milk-based fortifier powder can be a reasonable, noninferior cost alternative to standard fortifier in some settings ^[28].

None of this makes human-milk-based fortifier a poor choice — it makes it a genuine trade-off rather than a straightforward upgrade. A unit choosing to switch, or choosing not to, is making a defensible decision either way on current evidence, and that is worth saying explicitly in team discussions where a change in product is being proposed.

Individualised or "targeted" fortification — adjusting the dose to a baby's own milk composition — improves growth in the best current pooled analysis, but its effect on death, NEC and brain development remains "very uncertain" by the reviewers' own words ^[29]. This needs to be read alongside one sobering finding: a trial of a hands-on, ingredient-by-ingredient targeted approach had to stop early after five of the first 39 babies developed serious feeding intolerance ^[30]. Individualising fortification is a promising idea that has already shown it can go wrong when done informally — it is not something to improvise locally.

Practice point: do not present human-milk-based fortifier to parents as a proven way to prevent NEC — the honest answer is that it might help, according to smaller studies, but the best single trial found no difference.

Does fortification upset babies' tummies?

Partly, and this is measurable rather than assumed. In a trial comparing bovine colostrum with a standard fortifier, stools hardened as fortifier dose increased in *both* groups — confirming that harder stools really do track with fortifier dose, regardless of product, though less so with colostrum ^[31]. A Chinese cohort of 298 babies found no overall link between fortifier use and feeding intolerance, but did find that starting fortifier early, at full strength, or increasing it quickly were each independent risk factors ^[32].

Fortifier raises the osmolality (concentration) of milk, and commonly used medications add to this further ^[33], but this laboratory-level evidence about osmolality has not been directly linked to clinical outcomes in a trial ^[34,35]. The often-quoted 450 mOsm/kg safety ceiling is a sensible expert consensus figure, not one tested against real outcomes in a randomised trial.

Practice point: introducing fortifier gradually rather than at full strength, and watching stool pattern as a genuine clinical sign rather than an afterthought, is a low-cost, evidence-consistent habit even though no single trial has tested this practice directly.

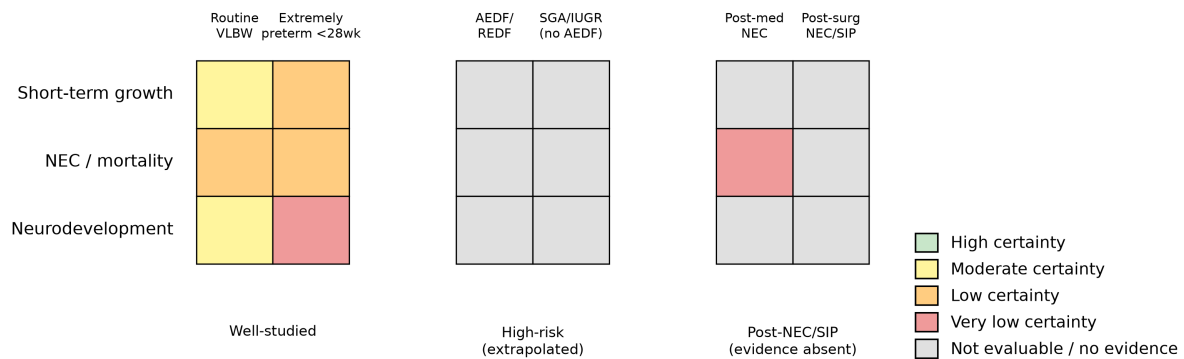


Figure 2. GRADE certainty by clinical scenario and outcome domain — colour shows certainty of evidence, not direction of effect.

Does fortification help the brain?

This is the question parents ask most, and it deserves the plainest possible answer: **no trial of any fortification variable — whether to fortify, which product, or how individualised — has yet shown a clear benefit to neurodevelopment.** A Danish trial following 214 babies to 6 years found no difference in IQ between those sent home on fortified versus unfortified mother's milk ^[36]. The N-forte fortifier-source comparison above found no 18-month developmental difference ^[23]. A trial of individualised fortification found real growth gains but admits its own developmental findings were mostly not statistically significant ^[37]. The N-forte team's planned 2- and 5.5-year follow-up, which could eventually give a firmer answer, has not yet been published ^[38].

Communicating with parents: this is often the single most useful thing a nurse can say clearly. *"Fortifying milk helps your baby grow a bit faster while in hospital. We don't have good evidence that it changes how your baby's brain develops later — and that's true whichever fortifier we use. Growing well now is still a good thing, but it isn't a guarantee about the future, and not growing quite as fast isn't a mark against your baby either."* Families frequently find this reassuring rather than disappointing, because it removes an implied pressure that faster growth now equals a better outcome later.

Going home: fortifying after discharge

The evidence here is consistently null on growth and development, with a different, more practical benefit showing through instead. A Danish trial found no growth difference at 12 months between babies sent home on fortified versus unfortified mother's milk ^[39], matching the 6-year IQ null above. A UK quality-improvement project found improved growth after introducing home fortifier use, though as an uncontrolled before-after study this cannot be attributed to the fortifier with confidence ^[40]. A US comparison found that continuing fortifier at home, rather than switching to formula-based top-ups, was linked to significantly higher rates of continued breast-milk feeding — with no difference in growth ^[41]. Note also that "nutrient-enriched formula" trials are a *different* intervention, for formula-fed babies, and should not be quoted as evidence about continuing human-milk fortifier at home ^[42].

Communicating with parents: frame continuing fortifier at home around supporting breastfeeding, not around an expected growth or developmental boost — that is what the evidence actually supports. *"If we suggest continuing the fortifier for a while after you go home, it's mainly to help you keep breastfeeding successfully — we don't have evidence that it will speed up growth or development beyond what your milk alone would do."*

What would change this picture

Three trials are worth watching. The N-forte trial's neurodevelopmental follow-up at 2 and 5.5 years, if published, will be the first properly powered long-term answer on fortifier source ^[38]. The ongoing three-arm MaxiMoM InForM trial (615 babies planned) is testing standard, targeted and blood-urea-nitrogen-adjusted fortification against an 18–24-month cognitive score — the first adequately sized trial with a hard developmental endpoint for

individualised fortification ^[43]. No trial of fortification timing in Doppler-abnormal or growth-restricted babies is currently known to be underway — the biggest gap identified here remains unfilled. Units auditing their own fortification practice do not need to wait for these trials: recording when fortification starts, which product is used, feed-intolerance events, and growth at discharge is achievable locally now, and would let a unit describe its own practice honestly even before the wider evidence base catches up.

Bringing it together: talking with parents

The communication points woven through this article share a common thread, and it is worth stating once, clearly, rather than leaving it scattered. Three habits make these conversations easier and more honest:

1. **Separate what is known from what is assumed.** "We fortify because it helps growth" is evidence-based. "We fortify because it will help your baby's development" is not currently supported and should not be said, however well-intentioned. 2. **Name uncertainty as uncertainty, not as a gap in the clinician's knowledge.** Parents generally cope well with "we don't have good research on this specific situation yet" when it is said plainly and paired with what *is* being done and why. What erodes trust is a confident-sounding answer that later turns out not to have been evidence-based. 3. **Revisit the conversation as circumstances change.** A baby who moves from Pathway A (Doppler-abnormal) to routine fortification once feeds are established, or who develops feeding intolerance after a fortifier change, is a natural moment to update parents on both what happened clinically and what is or isn't known about why.

Table 2 gives a starting form of words for the scenarios covered in this article; none of it needs reciting verbatim; it should be adapted to how each parent is taking the information in the moment.

Scenario	A form of words that fits the evidence
Explaining fortification generally	"Adding a fortifier to your milk gives your baby a bit more protein and energy than breast milk alone provides, which helps growth while in hospital."
Doppler-abnormal / growth-restricted baby	"We wait until your baby is tolerating full feeds well because babies like yours can have a more sensitive tummy — we don't yet have research on the best timing here specifically, so we're being cautious."
After NEC or bowel surgery	"We have evidence about safely restarting milk feeds. We don't have specific research on exactly when to add fortifier back in — your team will judge that step-by-step."
Choosing a fortifier product	"There are different types of fortifier. The evidence doesn't clearly favour one over another for safety, so we use our unit's standard product unless there's a specific reason to change."
Asked about long-term development	"Fortifying milk helps your baby grow a bit faster now. We don't have good evidence that it changes how your baby's brain develops later — growing well now is good, but it isn't a guarantee about the future."
Discharge planning	"If we suggest continuing fortifier for a while at home, it's mainly to help you keep breastfeeding successfully — not because it will speed up growth or development beyond what your milk alone would do."

Table 2. Suggested talking points by clinical scenario — adapt to how each family is taking the information.

What we know — and don't yet know — about milk fortification

A plain-language summary for talking with parents

	Does fortifying milk help my baby grow while in hospital? <i>Well-replicated across large trials and reviews</i>	YES — modestly, reliably
	Does it help my baby's brain develop in the long run? <i>No trial of any fortification approach has shown this yet</i>	NOT SHOWN
	Is cow's-milk-based or breast-milk-based fortifier clearly safer? <i>Smaller studies point one way; the largest trial found no difference</i>	GENUINELY UNCLEAR
	Is it safe to fortify after an abnormal blood-flow scan in pregnancy? <i>Care here is based on caution and experience, not a trial</i>	NO RESEARCH YET
	Is it safe to restart fortifier after bowel surgery (NEC)? <i>Decided case-by-case; no guideline covers this specific step</i>	NO RESEARCH YET
	Will continuing fortifier at home help growth or development? <i>May help keep breastfeeding going instead — a different benefit</i>	NOT SHOWN

Source: bmf-preterm-fortification evidence review, 50 PMID-verified studies. Not a substitute for individual clinical discussion.

Figure 3. Plain-language summary for parent conversations.

Conclusion

Fortification does what it was designed to do — it helps preterm babies grow a little faster in hospital — and that basic finding is well replicated. Almost everything built on top of that basic finding is less certain than routine practice implies: the ideal starting point, the best product, and above all whether any of it changes a baby's longer-term development. Being clear about this uncertainty, with colleagues and with parents, is not a weakness in practice — it is a more accurate and, families often find, more reassuring version of the truth. The evidence will move; the habit of describing it honestly, whatever it currently says, is worth keeping regardless.

Word count (main text, excluding key points, tables, figures and references): 2923 words. The full reference list (43 sources, all PMID-verified live against PubMed, with clickable links) is provided as a separate document.